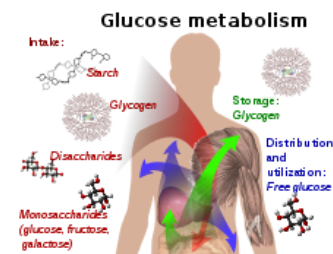


In other living organisms, other forms of fermentation can occur. The bacterium *Escherichia coli* can grow on nutrient media containing glucose as the sole carbon source.^[52] In some bacteria and, in modified form, also in archaea, glucose is degraded via the Entner-Doudoroff pathway.^[60]

Use of glucose as an energy source in cells is by either aerobic respiration, anaerobic respiration, or fermentation. The first step of glycolysis is the phosphorylation of glucose by a hexokinase to form glucose 6-phosphate. The main reason for the immediate phosphorylation of glucose is to prevent its diffusion out of the cell as the charged phosphate group prevents glucose 6-phosphate from easily crossing the cell membrane.^[61] Furthermore, addition of the high-energy phosphate group activates glucose for subsequent breakdown in later steps of glycolysis. At physiological conditions, this initial reaction is irreversible.

In anaerobic respiration, one glucose molecule produces a net gain of two ATP molecules (four ATP molecules are produced during glycolysis through substrate-level phosphorylation, but two are required by enzymes used during the process).^[62] In aerobic respiration, a molecule of glucose is much more profitable in that a maximum net production of 30 or 32 ATP molecules (depending on the organism) through oxidative phosphorylation is generated.^[63]

Click on genes, proteins and metabolites below to link to respective articles. ^[§ 1]



Glucose metabolism and various forms of it in the process. Glucose-containing compounds and isomeric forms are digested and taken up by the body in the intestines, including starch, glycogen, disaccharides and monosaccharides. Glucose is stored in mainly the liver and muscles as glycogen. It is distributed and used in tissues as free glucose.

Glycolysis and Gluconeogenesis edit (<http://www.wikipathways.org/index.php/Pathway:WP534>)

1. The interactive pathway map can be edited at WikiPathways: "GlycolysisGluconeogenesis_WP534" (<http://www.wikipathways.org/index.php/Pathway:WP534>).

Tumor cells often grow comparatively quickly and consume an above-average amount of glucose by glycolysis,^[64] which leads to the formation of lactate, the end product of fermentation in mammals, even in the presence of oxygen. This effect is called the Warburg effect. For the increased uptake of glucose in tumors various SGLT and GLUT are overly produced.^{[65][66]}

In yeast, ethanol is fermented at high glucose concentrations, even in the presence of oxygen (which normally leads to respiration but not to fermentation). This effect is called the Crabtree effect.

Energy source

Glucose is a ubiquitous fuel in biology. It is used as an energy source in organisms, from bacteria to humans, through either aerobic respiration, anaerobic respiration (in bacteria), or fermentation. Glucose is the human body's key source of energy, through aerobic respiration, providing about 3.75 kilocalories (16 kilojoules) of food energy per gram.^[67] Breakdown of carbohydrates (e.g., starch) yields mono- and disaccharides, most of which is glucose. Through glycolysis and later in the reactions of the citric acid cycle and oxidative phosphorylation, glucose is oxidized to eventually form carbon dioxide and water, yielding energy mostly in the form of ATP. The insulin reaction, and other mechanisms, regulate the concentration of glucose in the blood. The physiological caloric value of glucose, depending on the source, is 16.2 kilojoules per gram^[68] and 15.7 kJ/g (3.74 kcal/g), respectively.^[69] The high availability of carbohydrates from plant biomass has led to a variety of methods during evolution, especially in microorganisms, to utilize the energy and carbon storage glucose. Differences exist in which end product can no longer be used for energy production. The presence of individual genes, and their gene products, the enzymes, determine which reactions are possible. The metabolic pathway of glycolysis is used by almost all living beings. An essential difference in the use of glycolysis is the recovery of NADPH as a reductant for anabolism that would otherwise have to be generated indirectly.^[70]

Glucose supplies almost all the energy for the brain,^[71] so its availability influences psychological processes. When glucose is low, psychological processes requiring mental effort (e.g., self-control, effortful decision-making) are impaired.^{[72][73][74][75]} In the brain, which is dependent on glucose as the major source of energy, the glucose concentration is usually 4 to 6 mM (5 mM equals 90 mg / dL),^[35] but decreases to 2 to 3 mM when fasting.^[76] Confusion occurs below 1 mM and coma at lower levels.^[76]